Smart Thermally Actuating Textiles in Wearable Technology: A Lumbar Support Garment for Wheelchair Users

Audrey Percich, Julianna Abel, Mechanical Engineering, University of Minnesota

Background and Motivation

Problem:
- Pain - Lumbar compression from sitting and weak trunk strength.
- Mobility/Pressure Sores - Poor posture or slipping.
- Bulky Medical Equipment - See current market solutions.

Main Goal:
- Comfortable and adjustable postural and lumbar support assistance.



Common Pressure Sore Locations.
Boney Protrusions Locations for Sores

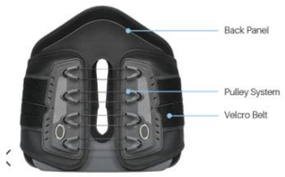
Current Market Products:



[1]



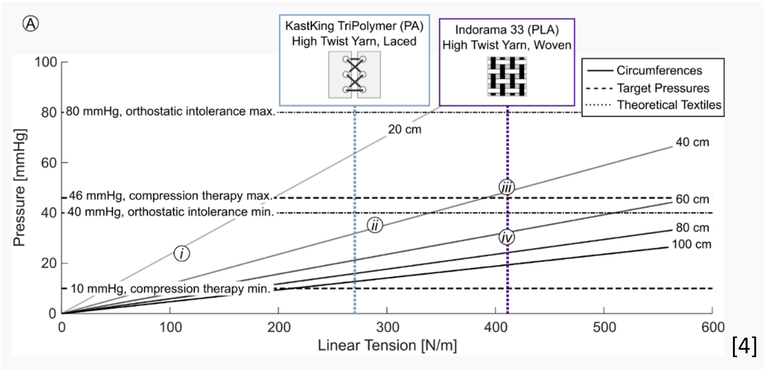
[2]



[3]

Linear Tension Requirements

Model
- Medical compression garments often target specific pressure outcomes.
- The linear tension model converts pressure to tension for proof of concept [4].
  - Assumes a rigid, cylindrical waist and overestimates pressure.



[4]

Anthropometrics
- Wheelchair user data informed the model.
- 86.63 cm waist circumference was selected from a research article [5].

Medical Metric for Semi-ridged LSO

$$25.95mmHg \rightarrow 3.459716kPa$$

$$r = \frac{86.63 * 10^{-3}}{2\pi} = 0.137876m$$

$$P = \frac{T}{r} \rightarrow T = P * r$$

$$T = 3459.716 * 0.137876$$
$$T = 477N/m$$

Medical Metric for Solid LSO

$$60mmHg \rightarrow 8kPa$$

$$r = \frac{86.63 * 10^{-3}}{2\pi} = 0.137876m$$

$$P = \frac{T}{r} \rightarrow T = P * r$$

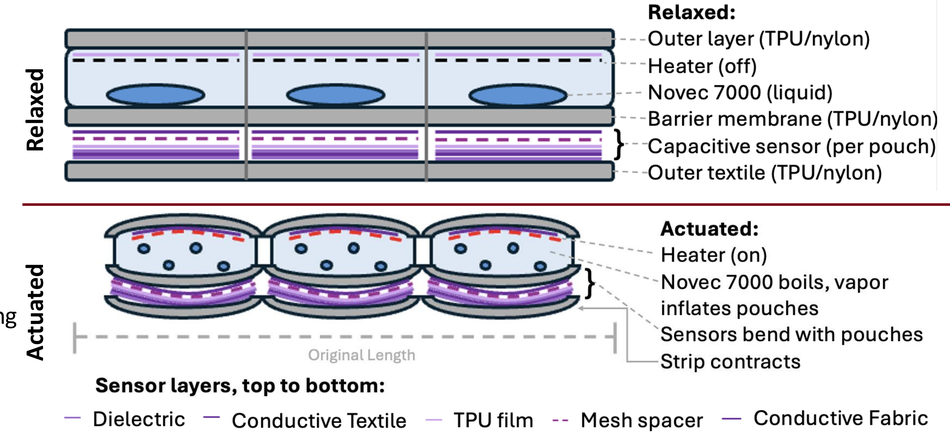
$$T = 8000 * 0.137876$$
$$T = 1103N/m$$

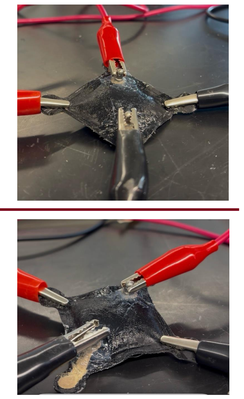
Manufacturing Smart Thermally Actuating Textile (STAT)

STATs Main Components [6]
- Heater
- Sensor
- Barrier Membrane

STAT Manufacturing
- Laser Cut
- Hot Press Bonding

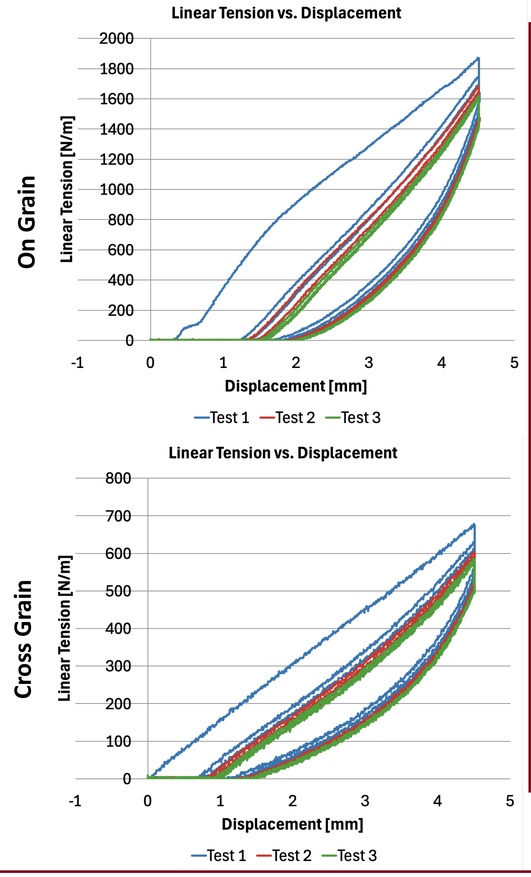
STAT Function
- Powered heater vaporizes working fluid
- STAT expands through thickness while reducing in width





Tensile Testing

TPU-Coated Nylon Tests:
- Woven material: on-grain and cross-grain directions.
- On-grain is about 3× stiffer than cross-grain.
- Soft TPU bonds limited tensile strength.
- TPU tensile strength: ~20 MPa.
- Estimated strength: 805 N (three layers, 40.25 mm²).
- Single-layer break force: ~268 N.
- Tested break force: 174.4 N



STAT's Tests:
- STAT tested 3x while fully actuated.
- Woven material: on-grain and cross-grain directions.
- On-grain is about 3× stiffer than cross-grain.
- Estimated strength: 805 N (three layers, 40.25 mm²).
- Single-layer break force: ~268 N.
- Tested break force: 173.8 N

On Grain

Cross Grain

GRAPHS HERE

Conclusion

Summary:
- Linear tension model indicates that STATs meet requirements for an LSO

Next Steps:
- Refine
  - Actual waist measurements
  - New geometry models
- Develop sensor-based control system
- Prototype

Bibliography

1. Freedom Gain – Lumbar Support Brace for sitting & recovery. Wheelchair88. 2025 May 19 [accessed 2025 Dec 6]. https://wheelchair88.sg/product/lumbar-support-brace/
2. Wheelchair back cushion lumbar support. Corner Home Medical. 2020 Feb 13 [accessed 2025 Dec 6]. https://cornermedical.com/product/wheelchair-back-cushion-lumbar-support/
3. Wheelchairs. Southwest Medical. [accessed 2025 Dec 6]. https://www.southwestmedical.com/Wheelchairs/Wheelchair-Positioning/Upper-Body-Positioning/JAY-Classic-Anterior-Trunk-Support/40074p?cid=1257
4. M. Andrews, S. Mantell, and J. Abel, " *Material Selection and Geometric Design for Few-filament Shape Memory Polymer Yarns for Force Generation*." *Advanced Materials Technologies* 11, no. 13 (2026): e02402. https://doi.org/10.1002/admt.202502402
5. Zwierzchowska, A., Głowacz, M., Batko-Szwaczka, A., Dudzińska-Griszek, J., Mostowik, A., Drozd, M., & Szewieczek, J. (2014). The Body Mass Index and Waist Circumference as Predictors Of Body Composition in Post CSCI Wheelchair Rugby Players (Preliminary Investigations). *Journal of human kinetics*, *43*, 191–198. https://doi.org/10.2478/hukin-2014-0105
6. Sanchez, V., Payne, C. J., Preston, D. J., Alvarez, J. T., Weaver, J. C., Atalay, A. T., ... & Walsh, C. J. (2020). Smart thermally actuating textiles. *Advanced Materials Technologies*, *5*(8), 2000383.